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Internal Q&A Knowledge Sharing Platform for BEL

AN INTERNSHIP REPORT

Submitted by

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Under the guidance of,

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BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND TECHNOLOGY (ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

GAIN MORE KNOWLEDGE
REACH GREATER HEIGHTS

PRESIDENCY UNIVERSITY

BENGALURU

APRIL 2026

PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

BONAFIDE CERTIFICATE

Certified that this report **Internal Q&A Knowledge Sharing Platform for BEL** is a bonafide work of **Likith R - 20221CSG0001,** who has successfully carried out the internship work and submitted the report for partial fulfilment of the requirements for the award of the degree of BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE AND TECHNOLOGY (ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING) during 2025-2026

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He was regular, punctual and his conduct was good during the period.

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LIKITH R

ABSTRACT

This project presents an Internal Knowledge Sharing and Q&A Platform for Bharat Electronics Limited (BEL), aimed at enhancing organizational knowledge management and collaboration. It addresses the challenge of fragmented information and repeated problem-solving across departments by providing a centralized, structured platform inspired by modern developer communities. The system enables users to post questions, contribute answers, and participate in discussions through a feature-rich interface that includes tagging, commenting, voting, and answer acceptance mechanisms. A reputation-based points system and leaderboard promote active participation while ensuring content quality. Role-based access control and moderation features maintain system integrity and enforce organizational policies. The application is built using a modular architecture, with a frontend developed in HTML, CSS, and JavaScript, a backend powered by Node.js and Express, and a MySQL relational database for efficient data management. RESTful APIs facilitate seamless communication between system components, ensuring scalability and maintainability. An audit trail mechanism tracks edit history, enhancing transparency and accountability. The system reduces redundancy, improves response time, and supports future enhancements like AI-based recommendations, real-time notifications, and enterprise integration.

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ABBREVIATIONS

Abbreviation	Full Form
API	Application Programming Interface
BEL	Bharat Electronics Limited
CSS	Cascading Style Sheets
CRUD	Create, Read, Update, Delete
DB	Database
ER	Entity Relationship
HTML	Hypertext Markup Language
HTTP	Hypertext Transfer Protocol
IDE	Integrated Development Environment
JS	JavaScript
JSON	JavaScript Object Notation
KMS	Knowledge Management System
MVC	Model-View-Controller
MySQL	My Structured Query Language
NASSCOM	National Association of Software and Service Companies
OS	Operating System
Q&A	Question and Answer
RAM	Random Access Memory
SDG	Sustainable Development Goals
SQL	Structured Query Language
UI	User Interface
UML	Unified Modeling Language
URL	Uniform Resource Locator

Chapter 1

INTRODUCTION

In large organizations like Bharat Electronics Limited (BEL), a significant amount of knowledge is generated on a daily basis through projects, problem-solving, and operational work. However, this knowledge is often not properly documented or shared across teams. Most of it stays with individuals or within specific departments, making it difficult for others to access when required.

During the internship, it was observed that employees often rely on emails, phone calls, or personal communication to resolve technical or work-related issues. While this works in the short term, it becomes inefficient over time because the same questions are asked repeatedly and solutions are not stored in a reusable format.

To address this issue, the project focuses on building an Internal Q&A Knowledge Sharing Platform, where employees can ask questions, share answers, and create a structured repository of knowledge. The idea is simple but powerful: instead of solving the same problem multiple times, solve it once and make it available to everyone.

1.1 Background

Knowledge management has been one of the main concerns of organizational success in particular in technical and engineering-based contexts such as BEL. Although there are highly qualified employees and experience, in most cases, the absence of a centralized system causes inefficiencies. As it could be observed through the internship, the majority of the knowledge transfer occurs informally. As an example, when an employee has a problem, he or she is likely to contact a person he or she knows and this person may have encountered the same problem in the past. This brings about dependence of individuals as opposed to systems. Research conducted by World Bank suggests that any organization that is able to manage and share knowledge will be doing well in terms of productivity and innovation. On the same note, studies on knowledge management indicate that collaborative platforms are instrumental in enhancing speed of resolving a problem as well as lessening redundancy. The necessity of having a structured and interactive platform is even more significant in the case of BEL due to its size and work division. The centralized Q&A system can be used to overcome the barriers between departments and make sure the knowledge gained is not lost over the time.

1.2 Statistics

One of the major reasons for developing this system is the amount of time employees spend searching for information. A study by McKinsey & Company found that employees spend nearly 20% of their workweek looking for internal information or identifying people who can help with specific tasks.

In large organizations, this leads to:

- Waste of productive time
- Delayed project completion
- Increased dependency on experienced employees

In the Indian context, especially in organizations like BEL, the situation is more noticeable due to hierarchical structures and distributed teams. Reports from NASSCOM highlight that digital collaboration tools can significantly improve efficiency and knowledge accessibility in such environments.

During the internship, informal observations also showed that similar technical questions were being asked multiple times across teams. This clearly indicates the absence of a reusable knowledge system.

1.3 Prior Existing Technologies

Before proposing this system, it is important to understand what was already in place.

The commonly used methods for knowledge sharing include:

- Email communication
- Internal documentation systems
- Team meetings and discussions
- Reports and shared folders

While these methods are useful, they come with several limitations. Emails, for example, are not searchable in a structured way and are often lost over time. Documentation systems may contain information, but they are usually static and not interactive.

Platforms like Stack Overflow have shown how effective a structured Q&A system can be. Features like voting, tagging, and answer acceptance help in identifying the most relevant and reliable solutions quickly. However, such platforms cannot be used directly within organizations like BEL due to confidentiality and security concerns. This creates a clear gap that an internal platform can fill.

1.4 Proposed Approach

1.4.1 Aim of the Project

The main aim of this project is to develop a web-based platform that allows employees to ask questions, share answers, and build a centralized knowledge base within BEL.

1.4.2 Motivation

The motivation for this project came from observing repeated problem-solving efforts and the lack of a proper system to store and retrieve solutions. Instead of relying on memory or personal networks, a structured platform can make knowledge accessible to everyone.

1.4.3 Proposed Approach

The system is designed as a web application with a simple and intuitive interface. Users can log in, post questions, provide answers, and interact with existing content.

1.4.4 Key features include:

- Posting and answering questions
- Voting system to identify useful answers
- Tagging for better organization
- Accepting the best answer
- User authentication and access control

The backend handles data processing and storage, while the frontend ensures a smooth user experience.

1.4.5 Applications of the Project

- Resolving technical queries within teams
- Helping new employees learn faster
- Reducing dependency on specific individuals
- Creating a long-term knowledge repository

1.4.6 Limitations of the Proposed Approach

- The system depends on active user participation
- Initial adoption may be slow
- Requires moderation to maintain content quality
- Limited scalability without advanced infrastructure

1.5 Objectives

- Behavior: To come up with an interactive user interface that will help the users to post, browse, and answer questions effectively.
- Analysis: To develop a systematic database scheme of storing and retrieving of questions, answers, and tags in the most efficient way.
- System Management: To deploy the backend functionality that manages the activities of the users, data flow and system performance efficiently
- Security: In order to achieve such safety access by use of authentication and authorization systems, internal data should be secured against unauthorized access.
- Deployment: The objective of the project was to make a working web application which can be tested and deployed in an organizational setting.

1.6 SDGs (Sustainable Development Goals)

Figure 1.1 SDG (Sustainable Development Goals

This project aligns with global development goals defined by the United Nations:

1. SDG 9: Industry, Innovation, and Infrastructure

   The project promotes innovation by introducing a digital knowledge-sharing system within an organization.

2. SDG 4: Quality Education

   It supports continuous learning by making knowledge easily accessible to employees.

3. SDG 8: Decent Work and Economic Growth

   By improving efficiency and reducing time wastage, the system contributes to better productivity.

1.7 Overview of the Internship report

- Chapter 1 introduces the study by discussing the background of knowledge management within large organizations like Bharat Electronics Limited (BEL). It highlights the challenges of traditional knowledge-sharing methods, presents supporting statistics, reviews existing practices, and explains the proposed approach of developing an Internal Q&A Knowledge Sharing Platform. It also outlines the objectives of the project and its alignment with Sustainable Development Goals (SDGs).
- Chapter 2 presents the literature survey, covering previous research on knowledge management systems, collaborative platforms, and organizational knowledge sharing. It analyzes existing technologies and identifies their limitations, particularly in terms of interactivity and security, thereby justifying the need for a dedicated internal Q&A platform.
- Chapter 3 describes the methodology followed in the project. It explains the step-by-step approach including problem analysis, requirement gathering, system design, implementation, and testing. The chapter also details the modular structure of the system and the tools and technologies used for development.
- Chapter 4 focuses on project management aspects, including the internship timeline, phase-wise development plan, risk analysis, and budget estimation. It highlights how the project was executed in a structured manner within the given time frame.
- Chapter 5 covers the analysis and design of the system. It includes functional and non-functional requirements, system architecture, block diagrams, flowcharts, module descriptions, database design, communication model, and overall system design considerations to ensure scalability and usability.
- Chapter 6 provides details about the hardware and software environment used for development. It discusses system requirements, development tools, code structure, and the testing approach through simulated user interactions to validate system performance.

- Chapter 7 presents the evaluation and results of the system. It includes test scenarios, testing methodology, observed results, and performance analysis. The chapter also highlights insights gained from testing and identifies areas for improvement.
- Chapter 8 discusses the social, legal, ethical, sustainability, and safety aspects of the project. It explains how the system impacts organizational collaboration, ensures secure data handling, promotes ethical usage, and supports sustainable practices.
- Chapter 9 concludes the report by summarizing the overall outcomes of the project. It highlights the effectiveness of the developed system and suggests future enhancements such as advanced search features, recommendation systems, scalability improvements, and enhanced security mechanisms.

Chapter 2

LITERATURE SURVEY

2.1 Literature Review

The concept of knowledge sharing within organizations has been widely studied over the years, especially with the rise of digital collaboration tools. Organizations have increasingly recognized that knowledge is not just an individual asset but something that should be shared, reused, and continuously improved.

One of the foundational works in this area is by **Alavi and Leidner (2001)**, who explained that knowledge management systems help organizations in creating, storing, and transferring knowledge effectively. Their study highlights that organizations with proper knowledge-sharing mechanisms tend to perform better in terms of productivity and innovation. This directly supports the idea behind developing an internal Q&A platform.

Another important perspective comes from research on collaborative systems, which shows that platforms enabling user interaction led to faster problem-solving. Systems where users can ask questions and receive answers from peers create a form of collective intelligence. This is exactly the principle used in platforms like Stack Overflow, where community participation helps in identifying the most accurate and useful solutions through voting and feedback mechanisms.

A study by McKinsey & Company (2012) emphasizes that employees spend a significant portion of their time searching for information or the right person to solve a problem. The study suggests that implementing social technologies within organizations can improve productivity by up to 20–25%. This clearly indicates the importance of having a structured platform for knowledge sharing.

In the Indian context, reports from NASSCOM highlight the growing need for digital transformation in workplaces. Many organizations are moving towards internal platforms that support collaboration, especially in large enterprises where communication gaps are common.

Traditional knowledge management systems, such as document repositories and intranet portals, have also been studied extensively. While they are useful for storing information, they often lack interactivity and real-time engagement. Users typically find it difficult to search for

specific solutions, and there is no mechanism to validate whether the information is still relevant or accurate.

Modern approaches focus on making knowledge systems more dynamic and user-driven. Features like tagging, voting, and answer acceptance have been shown to significantly improve the usability of such systems. These features not only help in organizing information but also ensure that the best solutions are easily visible to users.

However, most existing public platforms cannot be directly used within organizations like BEL due to confidentiality and security concerns. This creates a gap between available technology and organizational requirements. An internal Q&A platform addresses this gap by providing similar functionality while ensuring data privacy.

From the reviewed literature, it is clear that:

- Knowledge sharing improves organizational efficiency
- Interactive systems are more effective than static repositories
- Community-driven validation increases reliability of information
- There is a strong need for secure, internal platforms in large organizations

Based on these insights, the proposed system is designed to combine the strengths of existing Q&A platforms with the specific needs of an organization like BEL.

2.2 Summary

The literature review carried out for this project indicates that while several knowledge management and collaborative platforms exist, a fully integrated and organization-specific solution is still not commonly implemented in many traditional enterprises.

Most existing systems focus on individual aspects of knowledge sharing, such as document storage, communication, or discussion forums. However, they often fail to provide a complete, structured, and interactive solution that enables efficient problem-solving and long-term knowledge retention.

Traditional systems like document repositories and intranet portals primarily act as static

storage units. They lack dynamic features such as real-time interaction, answer validation, and structured categorization. As a result, users often face difficulties in retrieving relevant information quickly, leading to repeated queries and inefficiencies.

On the other hand, modern platforms such as Stack Overflow demonstrate the effectiveness of community-driven knowledge sharing. These platforms successfully integrate features like voting, tagging, and answer acceptance to improve content quality and accessibility. However, they are not suitable for internal organizational use due to concerns related to data privacy, security, and confidentiality.

Furthermore, studies from McKinsey & Company highlight that employees spend a significant portion of their time searching for information, which directly impacts productivity. While some organizations have started adopting internal collaboration tools, these systems often lack proper structure and scalability.

Another key observation is that many existing solutions do not provide a unified pipeline that combines:

- Knowledge creation
- Knowledge validation
- Knowledge storage
- Knowledge retrieval

This fragmentation reduces the overall efficiency and usability of such systems.

To address these gaps, this project proposes the development of an **Internal Q&A Knowledge Sharing Platform for BEL**, designed as a centralized and interactive system. The platform integrates question-answer mechanisms, tagging, voting, and user authentication into a single cohesive solution.

The system aims to:

- Minimize repeated problem-solving
- Improve knowledge accessibility

- Enhance collaboration among employees
- Build a scalable and reusable knowledge base

By doing so, the project provides a practical and efficient solution tailored to organizational needs, supporting long-term knowledge management and operational efficiency.

Table 1.1 Summary of Literature reviews

Reference	Focus Area	Key Findings	Accuracy/Performance	Limitations
Alavi & Leidner [1]	Knowledge Management Systems	Effective knowledge sharing improves organizational performance	High conceptual effectiveness	Lacks implementation details
McKinsey & Company [2]	Workplace productivity	Employees spend ~20% time searching for information	Significant productivity improvement possible	Not system-specific
Stack Overflow [3]	Community-driven Q&A	Voting and tagging improve answer quality and retrieval	Highly scalable and efficient	Not suitable for private organizational data
NASSCOM [4]	Digital collaboration in India	Increasing need for internal knowledge platforms	Improves workforce efficiency	Limited adoption in traditional enterprises
Bock et al. [5]	Knowledge sharing behavior	User participation depends on motivation and ease of use	Improves collaboration when adopted	Adoption challenges
Hansen et	Knowledge	Codification	Reduces redundancy	Requires

Reference	Focus Area	Key Findings	Accuracy/Performance	Limitations
al. [6]	strategy	strategy helps reuse knowledge effectively		structured systems
Grudin [7]	Groupware systems	Collaboration tools fail if not user-friendly	Moderate success rate	Poor usability leads to failure
Dalkir [8]	Knowledge lifecycle	Knowledge must be created, stored, and shared systematically	Improves lifecycle management	Needs integrated systems
O'Reilly [9]	Web-based platforms	Interactive platforms outperform static systems	High engagement levels	Requires active users
Tiwana [10]	Knowledge portals	Centralized systems improve accessibility	Better information flow	Implementation complexity

Chapter 3

METHODOLOGY

This chapter explains the approach followed for developing the Internal Q&A Knowledge Sharing Platform for BEL. The development process was carried out in a structured manner, starting from understanding the problem to designing, implementing, and testing the system. The methodology focuses on building a practical and usable solution rather than just a theoretical model. The entire workflow is divided into stages such as requirement analysis, system design, development, testing, and evaluation. The goal was to ensure that the system is not only functional but also easy to use and adaptable within an organizational environment. Unlike traditional knowledge systems, this project emphasizes interaction, real-time usage, and structured data handling. The approach ensures that the system can handle multiple users, manage content efficiently, and provide reliable outputs.

3.1 Research Method

The development of this project followed a step-by-step approach combining observation, design thinking, and implementation.

Problem Analysis:

During the internship, it was observed that employees often rely on informal communication like emails or direct contact to solve problems. This leads to repeated questions, lack of documentation, and inefficiency.

Requirement Gathering:

Based on these observations, the need for a centralized platform was identified. The system should allow users to ask questions, provide answers, and access previously solved queries easily.

System Design:

A web-based architecture was planned, consisting of frontend, backend, and database layers.

Implementation:

The system was developed in modules such as user authentication, question management, answer handling, and voting system. Each module was implemented separately and later integrated.

Testing and Evaluation:

The system was tested with sample data to ensure proper functioning of all features. Different scenarios were checked, such as posting questions, answering, and retrieving information.

Result Analysis:

The final system was evaluated based on usability, response time, and correctness of data handling. Improvements were identified for future enhancements.

This step-by-step approach helped in maintaining clarity and avoiding confusion during development.

3.2 Module Description

The system is divided into multiple modules, each responsible for a specific function. These modules work together to create a complete knowledge-sharing platform.

3.3 User Management Module

This module handles user registration, login, and authentication. Each user is assigned a unique identity, allowing the system to track their activity such as questions posted, answers given, and interactions.

Basic authentication mechanisms are implemented to ensure that only authorized users can access the platform. This is important since the system is designed for internal organizational use.

3.4 Question Management Module

This module allows users to post questions related to their work or technical issues. Each question includes a title, description, and tags for categorization.

The tagging system plays an important role in organizing content, making it easier for other users to find relevant questions.

3.5 Answer Management Module

Once a question is posted, other users can provide answers through this module. Answers are stored in relation to their respective questions.

This module ensures that multiple answers can be submitted, allowing users to choose the most suitable solution.

3.6 Voting and Acceptance Module

This module introduces interaction and quality control into the system. Users can upvote or downvote answers based on usefulness.

Additionally, the person who asked the question can mark one answer as the "accepted answer," helping future users quickly identify the best solution.

3.7 Data Storage Module

All data related to users, questions, answers, and interactions is stored in a structured database. The database ensures:

- Efficient storage
- Fast retrieval
- Data consistency

It acts as the backbone of the system, maintaining all records securely.

3.8 Workflow and Data Flow

The system follows a simple and logical workflow:

1. User logs into the platform
2. User posts a question or searches for existing ones
3. Other users view the question and provide answers
4. Answers can be voted on by users
5. The best answer is accepted by the question owner
6. All data is stored and can be accessed later

This workflow ensures smooth interaction and continuous knowledge building within the system.

3.9 Tools and Technologies Used

The project uses commonly available tools and technologies:

- Frontend: HTML, CSS, JavaScript
- Backend: Node.js

- Database: MySQL
- Development Tool: Visual Studio Code
- Browser: For testing and execution

These tools were chosen because they are easy to use, widely supported, and suitable for building scalable web applications.

3.10 Summary

This chapter was used to describe the approach used to create the Internal Q&A Knowledge Sharing Platform. The system is easy to maintain, extend and understand with the help of the modular approach. The system allows sharing of knowledge in an organization by integrating both structured data processing and interaction to offer a viable solution. The approach makes sure that it is not only the technically sound platform it is also practical in the real-life context.

Chapter 4

PROJECT MANAGEMENT

This chapter explains how the project was planned and executed during the internship period at Bharat Electronics Limited (BEL). Even though the project was carried out individually, a structured approach was followed to ensure smooth progress and timely completion.

The project management process mainly focused on three aspects: planning the timeline, identifying possible risks, and estimating the resources required. This helped in keeping the development process organized and avoiding unnecessary delays.

4.1 Internship Timeline

The internship was conducted at PDIC, Bharat Electronics Limited (BEL), Bengaluru, over a period of three months. It was carried out on a full-time basis during working hours from 9:00 AM to 5:00 PM, following the organization's standard schedule.

Internship Details:

- Internship Period: 05 January 2026 – 06 April 2026
- Duration: Three Months
- Internship ID: PT-717
- Location: PDIC, BEL, Bengaluru
- Working Hours: 9:00 AM – 5:00 PM

The project development was divided into phases to ensure clarity and proper execution of tasks.

Phase-wise Timeline:

Phase 1: Initial Understanding and Requirement Analysis (Week 1–2)

The first few weeks were spent understanding the working environment at BEL and identifying the challenges in knowledge sharing. Based on observations and discussions, the requirements for the system were defined.

Phase 2: System Design (Week 3–4)

In this phase, the overall structure of the system was planned. This included designing the database, defining modules, and selecting appropriate technologies for development.

Phase 3: Backend Development (Week 5–6)

The backend logic of the application was developed, including API creation, database integration, and user authentication features.

Phase 4: Frontend Development (Week 7–8)

The user interface was designed and implemented. Focus was given to making the system simple and easy to use so that employees can interact with it without difficulty.

Phase 5: Integration and Testing (Week 9–10)

All modules were integrated, and the system was tested with different use cases to ensure proper functionality and performance.

Phase 6: Finalization and Documentation (Week 11–12)

The final stage involved refining the system, fixing minor issues, and preparing the internship documentation and report.

This phased approach ensured steady progress and helped in completing the project within the given duration.

4.2 Risk Analysis

Every project comes with its own set of risks, even if everything looks fine on paper. Identifying these risks early helped in managing them effectively during development.

Table 4.1: Project Risk Assessment and Mitigation Plan

Risk	Impact	Probability	Mitigation Strategy
Low user adoption	High	Medium	Keep UI simple and easy to use
Data inconsistency	Medium	Medium	Use proper validation and structured database
Security issues	High	Low	Implement login and access control
Integration issues	Medium	Medium	Test each module before combining
Time constraints	High	Medium	Follow timeline and prioritize core features
Bugs and errors	Medium	High	Perform regular testing and debugging

By addressing these risks during development, the system was made more stable and reliable.

4.3 Internship Budget

The project primarily involves software development, so the overall financial requirement is minimal. However, certain indirect costs and resource considerations are involved in the development and deployment of the system.

Estimated Budget Breakdown

Table 4.2: Cost Estimation and Resource Allocation for the Project

Component	Description	Cost
Development Tools	Visual Studio Code (open-source IDE)	Free
Programming Languages	HTML, CSS, JavaScript, Node.js	Free
Database	MySQL (open-source)	Free
System Hardware	Laptop/Desktop used for development	Existing Resource
Internet Usage	Required for development, testing, and research	Low
Hosting/Deployment	Optional (for real-time deployment on server/cloud)	Minimal
Maintenance	Future updates and system improvements	Variable

Total Estimated Cost: Low (primarily dependent on deployment and scaling requirements)

Additional Considerations

Although the development cost is minimal, deploying the system in a real organizational environment may involve additional expenses such as:

- Server infrastructure or cloud services
- Security enhancements
- Maintenance and support
- User training and onboarding

Chapter 5

Analysis and Design

This chapter focuses on the analysis and design of the Internal Q&A Knowledge Sharing Platform developed for BEL. The goal of this phase was to clearly define system requirements and design a structured architecture that ensures smooth functionality, scalability, and ease of use.

The design process emphasizes modularity, allowing different components of the system to function independently while still working together as a complete application.

5.1 Requirements

The requirements of the system are divided into functional and non-functional requirements.

Functional Requirements

- Registration of user and login
- Posting and managing questions
- Answering questions
- Voting on answers
- Accepting the best answer
- Tagging and categorizing questions
- Searching and filtering content

Non-Functional Requirements

- Simple and user-friendly interface
- Fast response time
- Data consistency and reliability
- Secure authentication and access control
- Ability to handle multiple users

5.2 Block Diagram

The system consists of three main components:

This layered architecture ensures separation of concerns and supports efficient system performance**.** This structured flow ensures smooth communication and efficient system operation.

- Frontend (User Interface)
- Backend (Server and API)
- Database (Data Storage)

The proposed system is designed using a **three-tier architecture**, which separates the application into distinct layers. This separation improves maintainability, scalability, and overall system performance.

Presentation Layer (Frontend

This layer is responsible for handling all user interactions. It includes the graphical user interface through which users can log in, post questions, view answers, and interact with the system. The frontend is designed to be simple and intuitive so that even non-technical users can navigate the platform easily. It communicates with the backend through API calls and displays data dynamically.

Application Layer (Backend)

The backend acts as the core processing unit of the system. It handles all business logic such as user authentication, question management, answer handling, and voting mechanisms. This layer ensures that all user requests are processed correctly and securely. It also manages communication between the frontend and the database.

Data Layer (Database)

The data layer is responsible for storing and managing all system data, including user details, questions, answers, and tags. A structured relational database is used to ensure efficient storage and quick retrieval of information. Proper indexing and relationships are maintained to optimize performance.

System Flow Explanation

When a user performs an action, such as posting a question, the request is sent from the frontend to the backend. The backend processes the request, interacts with the database to

store or retrieve data, and sends the response back to the frontend.

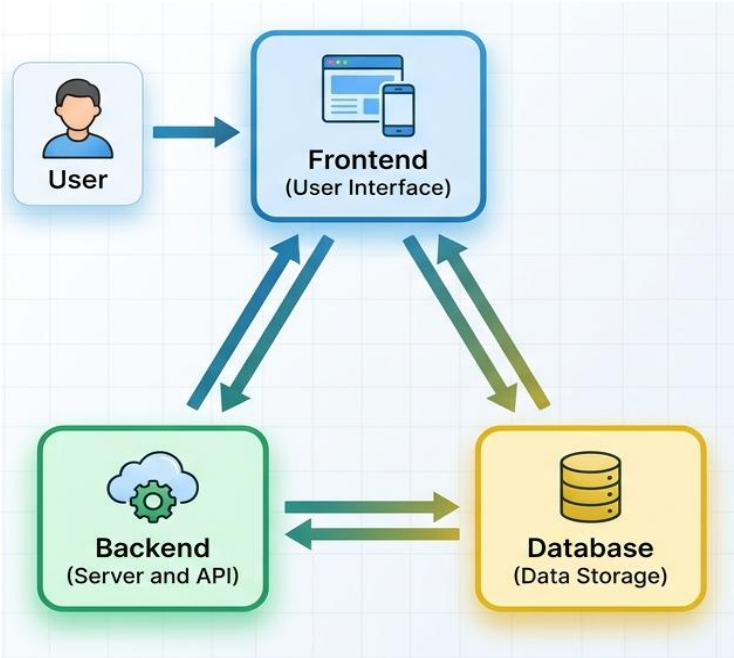

Figure 5.1: Block Diagram / System Architecture

5.3 System Flow Chart

The system follows a logical and sequential flow to ensure smooth interaction between users and the platform.

Initially, the user logs into the system using valid credentials. Once authenticated, the user is directed to the dashboard, where they can either post a new question or browse existing ones. When a question is posted, it is stored in the database along with relevant metadata such as user ID, timestamp, and tags.

Other users can view the question and submit answers based on their knowledge. These answers are stored and displayed in real time. Users can then evaluate answers by voting, which helps in identifying the most useful responses.

The user who posted the question has the ability to mark one answer as the accepted solution. This feature improves clarity for future users who encounter similar problems. All interactions are recorded and stored, creating a continuously growing knowledge base

. This flow ensures that information is not lost and can be reused effectively.

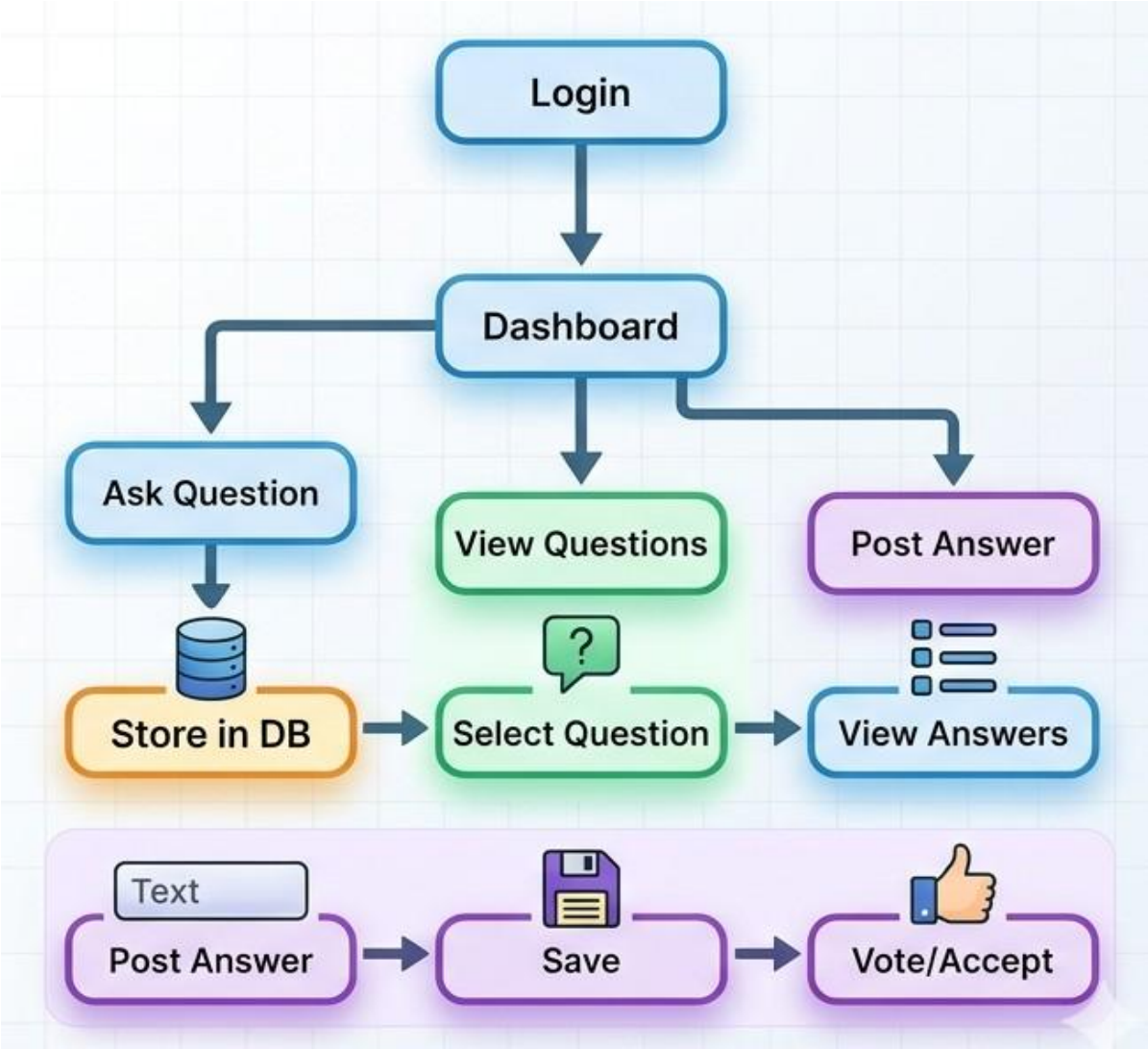

Figure 5.2: Flow Chart of the System

5.4 System Modules

The system is divided into multiple modules, each responsible for a specific function. This modular design improves clarity, maintainability, and scalability.

User Module

This module manages user-related operations such as registration, login, and profile handling. It ensures that only authorized users can access the system and interact with its features.

Question Module

This module allows users to create, edit, and view questions. It handles the storage and retrieval of question data and ensures proper categorization through tags.

Answer Module

The answer module manages the submission and display of answers. It links each answer to its corresponding question and ensures proper organization of responses.

Voting Module

This module allows users to vote on answers, helping in identifying the most useful and relevant solutions. It plays a key role in maintaining the quality of content within the platform.

Database Module

The database module is responsible for storing all system data in a structured format. It ensures data consistency, integrity, and efficient retrieval.

5.5 Database Design (Domain Model)

The database design is a crucial part of the system, as it determines how efficiently data can be stored and retrieved.

Entities

- User: Contains user information such as ID, name, email, and password
- Question: Stores details of questions including title, description, tags, and timestamps
- Answer: Contains answers linked to specific questions
- Tag: Used for categorizing and organizing questions

Relationships

- A user can post multiple questions and answers
- A question can have multiple answers
- Each answer is associated with one question
- Tags are linked to questions for better categorization

Design Considerations

The database is normalized to reduce redundancy and improve consistency. Proper indexing is used to enhance query performance. Relationships between tables are maintained using foreign keys, ensuring data integrity.

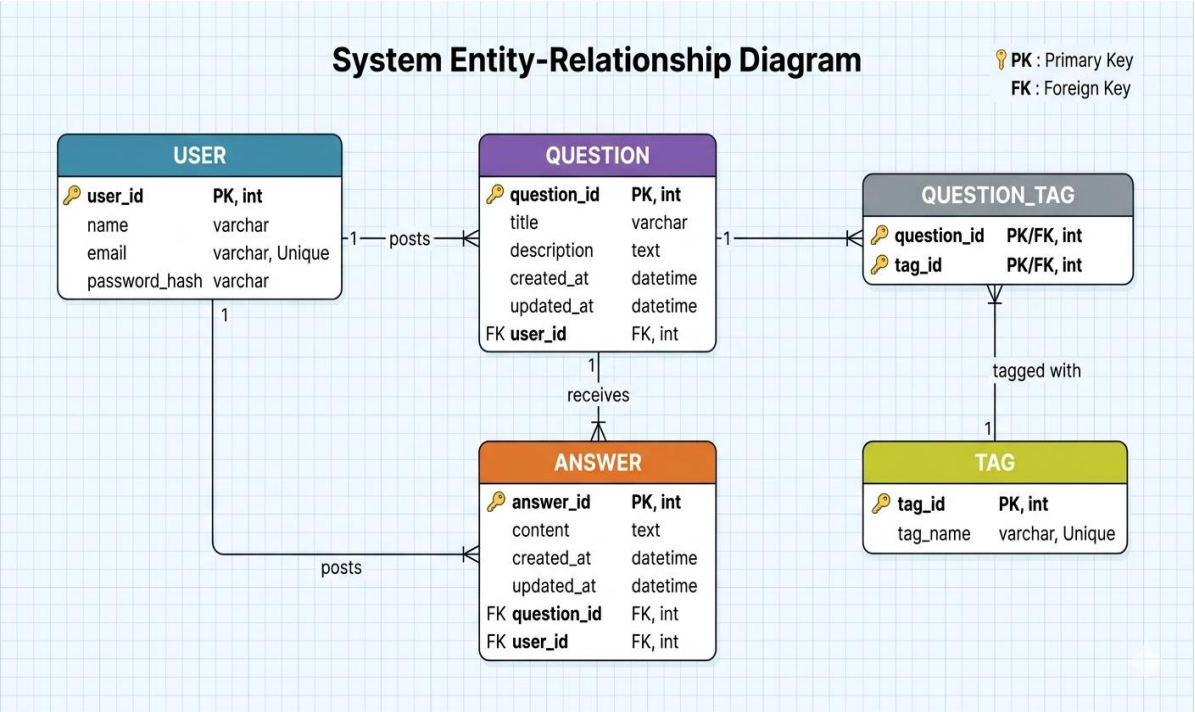

Figure 5.3: System ER Diagram of the System

5.6 Communication Model

The system follows a **client-server communication model**, which is widely used in web applications. When a user interacts with the system, the request is sent from the client (browser) to the server. The backend processes this request using APIs and performs necessary operations such as data retrieval or storage. The database is accessed only through the backend, ensuring security and control over data operations.

Once the processing is complete, the server sends a response back to the client, which is then displayed to the user.

This communication is carried out using HTTP protocols, ensuring reliable and standardized data exchange. This model ensures that the system remains secure, scalable, and easy to

manage.

5.7 Functional Design

The functional design of the system defines the core operations that enable effective knowledge sharing within the organization.

It focuses on how users interact with the platform and how different features support the overall objective of reducing repetitive queries and improving accessibility of information.

The system begins with user authentication, where each user logs in using valid credentials. This ensures that only authorized personnel can access the platform, maintaining data privacy within the organization.

Once logged in, users are provided with a dashboard that serves as the central interface. From here, users can perform multiple actions such as posting new questions, browsing existing discussions, and interacting with answers. The question posting feature allows users to clearly describe their issues, along with relevant tags that help categorize the content.

The answer management functionality enables other users to respond to questions based on their expertise or prior experience. Multiple answers can be provided for a single question, encouraging collaborative problem-solving.

To ensure quality control, a voting mechanism is implemented. Users can upvote or downvote answers, which helps in ranking them based on usefulness. This ensures that the most relevant and helpful answers are easily visible.

Additionally, the answer acceptance feature allows the question owner to mark one response as the final solution. This reduces ambiguity and helps future users quickly identify the correct answer.

The system also includes search and filtering capabilities, allowing users to find relevant questions based on keywords or tags.

This significantly reduces the time spent searching for information. Overall, the functional

design ensures that the platform is interactive, user-driven, and efficient in managing organizational knowledge.

5.8 Operational Design

The operational design focuses on how the system performs during real-time usage and how it handles multiple user interactions efficiently.

The platform operates in a real-time environment, where user actions such as posting questions, submitting answers, or voting are processed instantly.

This ensures a smooth and responsive experience for all users. The system is designed to support concurrent users, meaning multiple users can access and interact with the platform simultaneously without affecting performance. This is achieved through efficient request handling and optimized backend processing.

Each user request follows a structured flow: the request is received by the server, processed using backend logic, and the result is retrieved from or stored in the database. The response is then sent back to the user interface without noticeable delay.

To maintain system reliability, error handling mechanisms are implemented. These include validation of user inputs, handling invalid requests, and displaying appropriate error messages. This prevents system crashes and ensures smooth operation.

The system also ensures data consistency by managing database transactions properly. For example, when multiple users interact with the same question, the system ensures that updates are accurately reflected without conflicts.

In addition, basic security measures such as authentication and controlled access are implemented to protect sensitive organizational data.

The operational design ensures that the system remains stable, efficient, and reliable under different usage conditions.

Chapter 6

Hardware, Software and Simulation

The chapter identifies the tools and technical environment in the development of the Internal Q&A Knowledge Sharing Platform. This is a software-based system rather than one that relies on hardware unlike the hardware intensive projects, which lay emphasis on web technologies and database management.

But in the case of software-based system, the hardware, development tools and testing techniques are of great importance to make sure that there are smooth performance, scalability and usability. In this chapter, the author gives a clear picture of the infrastructure and procedures of building and validating the system.

6.1 Hardware

Even though the project is not associated with any particular hardware that requires either sensors or embedded systems, a stable computing environment will be required to support the project with its development, testing, and deployment.

Development Hardware Environment

The following setup was used to develop the system by using a regular computer:

- Processor: Intel i5 (or any other similar) and capable of multitasking several processes.
  • RAM: 8GB or more, which will guarantee the efficient use of development tools and local servers.
- Storage Adequate source code, database, and dependencies storage.
- Operation system: Windows/Linux based system.

The selected hardware setup will allow the development environment to perform such functions as a local server, database querying, and code debugging, without facing performance problems.

The chosen hardware configuration ensures that the development environment can handle tasks such as running a local server, executing database queries, and debugging code without performance issues.

User-Side Hardware Requirements

Since the platform is web-based, the end-user does not require any special hardware.

The system can be accessed through:

- Desktop or laptop systems
- Any device with a modern web browser
- Stable internet connection

This flexibility makes the system highly accessible and suitable for deployment within an organizational environment like BEL.

Hardware Considerations

While the project does not depend on advanced hardware, certain considerations were taken into account:

- Ensuring compatibility across different devices
- Maintaining performance even on moderate system configurations
- Avoiding hardware dependency to improve scalability

Overall, the minimal hardware requirement is one of the strengths of the system, as it reduces cost and simplifies deployment.

6.2 Software Development Tools

The system will be developed based on a frontend, backend and database combination. The selection of the tools was determined by the ease of use, the flexibility of the tools, and compatibility with the contemporary web development practices.

Frontend Technologies

- **HTML (Hypertext Markup Language):**
  Used for structuring the web pages and defining the layout of content.
- **CSS (Cascading Style Sheets):**
  Responsible for styling the user interface, including colors, layouts, and responsiveness.

- **JavaScript:**
  Enables dynamic behavior such as form validation, interactive elements, and asynchronous data fetching.

The frontend is designed to be simple and user-friendly, ensuring that users can easily navigate through the platform.

Backend Technologies

- **Node.js:**
  Used for server-side development, allowing the system to handle multiple requests efficiently.
- **Express.js**:
  Provides a framework for building APIs and managing routes, simplifying backend development.

The backend is responsible for processing user requests, handling authentication, and managing communication with the database.

Database

- **MySQL:**
  A relational database management system used to store structured data such as user details, questions, answers, and tags.

The database design ensures:

- Efficient data storage
- Fast retrieval of information
- Data consistency and integrity

Development Environment

- **Visual Studio Code:**
  Used as the primary code editor due to its simplicity and powerful extensions.
- **Web Browser (Google Chrome):**
  Used for testing and debugging the application.
- **Local Server Setup:**
  The application was run locally during development for testing purposes.

Tool Selection Justification

The tools were chosen based on the following factors:

- Open-source availability
- Ease of learning and implementation
- Strong community support
- Compatibility with modern web development standards

This combination ensures that the system is both efficient and easy to maintain.

6.3 Software Code

The system is implemented using a modular coding approach, where each functionality is handled by separate components. This improves readability, maintainability, and scalability of the code.

Code Structure

The application is divided into:

- Frontend Code: Handles user interface and interactions
- Backend Code: Manages logic, APIs, and data processing
- Database Queries: Handles data storage and retrieval

This separation ensures that each part of the system can be developed and modified independently.

Key Functional Modules in Code

- Authentication Module:
  Handles user registration and login using secure validation methods.
- Question Handling Module:
  Manages creation, retrieval, and display of questions.
- Answer Handling Module:
  Handles submission and storage of answers linked to questions.
- Voting Module:
  Allows users to evaluate answers, improving content quality.
- Search Module:
  Enables users to find relevant questions using keywords or tags.

6.4 Simulation

The project is a web-based application and therefore traditional methods of simulation applied in hardware or embedded system cannot be utilized. Nevertheless, the system was tested and checked by experimenting it and simulating user interactions.

Testing Environment

The application was tested in a local environment using:

- Sample user data
- Simulated scenarios involving multiple users
- Different types of inputs and edge cases

Simulation of User Interactions

Various real-world scenarios were simulated to evaluate system performance:

- Multiple users posting questions simultaneously
- Users answering and voting on questions
- Searching for existing content
- Handling incorrect or invalid inputs

This helped in understanding how the system behaves under different conditions.

Performance Evaluation

The system was evaluated based on:

- Response time for user requests
- Accuracy of data storage and retrieval
- System stability under multiple interactions

Error Handling and Validation

Special attention was given to:

- Handling invalid inputs
- Preventing duplicate data entries
- Ensuring proper error messages for users

Outcome of Simulation

The testing process confirmed that:

- The system performs efficiently under normal usage conditions
- Data is handled accurately and consistently
- The platform is stable and user-friendly

Chapter 7

Evaluation and Results

This chapter provides the assessment of the Internal Q&A Knowledge Sharing Platform that was created in the course of the internship. This evaluation will be done in order to check whether the system is fulfilling its intended purpose and it will be effective in various scenarios of use.

Various functional and user-based scenarios were used to test the system to determine its reliability, correction and usability. The analysis is not based on theoretical validation but on the practical implementation and interaction in real time.

7.1 Test Scenarios (Test Points)

To ensure proper functionality, multiple test scenarios were considered based on real-world usage of the platform.

User Authentication

- Valid login credentials
- Invalid login attempts
- New user registration

Question Management

- Posting a new question
- Editing and viewing questions
- Retrieving previously posted questions

Answer Management

- Submitting answers to a question
- Displaying multiple answers
- Linking answers correctly to questions

Voting System

- Upvoting and downvoting answers
- Updating vote count dynamically

Answer Acceptance

- Marking an answer as accepted
- Ensuring only one accepted answer per question

Search Functionality

- Searching questions using keywords
- Filtering questions using tags

These scenarios were selected to cover all major functionalities of the system.

7.2 Test Plan

The testing process was carried out in a structured manner to validate both functionality and performance of the system.

Testing Approach

- Manual testing of all features
- Functional testing of each module independently
- Integration testing after combining modules

Testing Environment

- Local system using browser-based execution
- Sample dataset representing real-world usage
- Multiple test cases for different scenarios

Steps Followed

1. Identify functionality to be tested
2. Execute test cases with valid and invalid inputs
3. Observe system behavior and outputs
4. Record results and identify issues
5. Fix errors and retest

This systematic approach ensured that all parts of the system were tested thoroughly.

7.3 Test Results

The results obtained from testing indicate that the system performs as expected under normal usage conditions.

Key Observations

- User authentication works correctly with proper validation
- Questions and answers are stored and retrieved accurately
- Voting mechanism updates in real time
- Accepted answers are correctly highlighted
- Search functionality returns relevant results

Performance

- The system responds quickly to user actions
- No major delays were observed during testing
- Data consistency was maintained across all operations

Error Handling

- Invalid inputs were handled appropriately
- Proper error messages were displayed
- System remained stable during incorrect operations

Overall, the system successfully passed all major test cases and demonstrated reliable performance.

7.4 Insights

The assessment exercise was able to give a number of valuable insights on the performance and usability of the system.

Among the most important observations, it is important to note that organized Q&A platform will greatly enhance the availability of knowledge in contrast to the traditional forms of communication.

The users can be able to find the solutions very fast without having to be involved in recurrent interactions.

The voting and acceptance of the answer's options are important in ensuring that the contents are good. These mechanisms make sure that the information of the best use is easily recognizable.

The other thing that I learned is the significance of user interface design. The interface should be simple and clean to enhance the level of interaction and less confusion among users; hence, making the system more effective.

The areas that might be improved in future were also identified in the testing process and included:

- Enhancing search accuracy
- Adding recommendation features
- Enhancing the scalability with bigger datasets.

In general, the system achieves its purposes and offers a viable knowledge sharing system in an organization.

Chapter 8

Social, Legal, And Ethical Aspects

This chapter talks about the overall effect of the Internal Q&A Knowledge Sharing Platform besides the technical implementation of the system. It points out the impact of the system on organizational culture, accountable use of data and compatibility with feasible and ethical practices. Although this is a software-based project, these are still relevant factors to consider so that the system becomes reliable, secure and can be deployed in the real-world scenario within an organization such as BEL.

8.1 Social Aspects

The platform contributes positively to the internal work environment by improving communication and collaboration among employees.

Traditionally, knowledge sharing within organizations often depends on informal communication methods such as emails or direct interactions. This can lead to repeated queries, delays in problem-solving, and uneven distribution of knowledge.

The proposed system creates a centralized platform where employees can ask questions and share solutions. This encourages a **collaborative culture**, where knowledge is not limited to individuals but is accessible to everyone.

It also helps in:

- Reducing dependency on specific individuals
- Improving efficiency in resolving issues
- Promoting continuous learning within the organization

Overall, the system strengthens teamwork and enhances productivity.

8.2 Legal Aspects

Since the platform is designed for internal use within BEL, data privacy and security are important considerations.

The system ensures that:

- Only authorized users can access the platform
- Internal organizational data is not exposed externally
- User information is stored securely

Basic authentication mechanisms are implemented to prevent unauthorized access. Additionally, the system is designed to comply with organizational policies regarding data usage and confidentiality.

While the project does not directly deal with external legal frameworks, it emphasizes responsible handling of internal data.

8.3 Ethical Aspects

Ethical considerations focus on how the system is used and how it impacts users.

The platform promotes ethical knowledge sharing by encouraging users to provide accurate and relevant information. Since answers are visible to all users, there is a level of accountability in the content being shared.

Key ethical considerations include:

- Avoiding misleading or incorrect information
- Respecting intellectual contributions of users
- Ensuring fair use of shared knowledge

The voting and answer acceptance mechanisms help in maintaining content quality by highlighting the most useful responses.

8.4 Sustainability Aspects

The system is designed to be sustainable in terms of both resource usage and long-term applicability.

Since it is a software-based solution:

- It does not require physical resources or hardware components
- It minimizes paper-based documentation
- It reduces time and effort spent on repeated problem-solving

By creating a reusable knowledge base, the platform ensures that information is preserved and can be accessed in the future, reducing redundancy.

Additionally, the use of open-source technologies makes the system cost-effective and adaptable for future enhancements.

8.5 Safety Aspects

The safety aspect mainly focuses on data security and system reliability.

The system incorporates:

- User authentication to restrict access
- Controlled data handling through backend validation
- Error handling to prevent system failures

Although there is no physical safety risks involved, ensuring the protection of user data and system stability is critical.

Proper handling of user inputs and secure storage of data reduce the chances of system misuse or data loss.

Chapter 9

Conclusion and Future Enhancements

The Internal Q&A Knowledge Sharing Platform developed during the internship at Bharat Electronics Limited (BEL) successfully addresses the challenges associated with traditional knowledge-sharing methods within an organization.

The project focused on creating a centralized and structured platform where employees can ask questions, share answers, and access previously solved problems. By replacing informal and scattered communication methods, the system improves efficiency, reduces redundancy, and ensures that knowledge is retained within the organization.

Throughout the development process, a systematic approach was followed, including requirement analysis, system design, implementation, and testing. Each phase contributed to building a functional and reliable application that meets the intended objectives.

The platform incorporates essential features such as user authentication, question and answer management, voting mechanisms, and answer acceptance. These features not only enhance usability but also ensure that the most relevant and accurate information is easily accessible to users.

The evaluation of the system demonstrated that it performs effectively under normal usage conditions. The system is capable of handling multiple users, maintaining data consistency, and providing quick responses. The testing process confirmed the reliability and correctness of the implemented functionalities.

In addition to technical aspects, the project also considered social, ethical, and organizational factors. The system promotes collaboration, encourages knowledge sharing, and supports a more efficient working environment.

Although the current implementation meets the core requirements, there is scope for further enhancement. Future improvements may include advanced search mechanisms, recommendation systems, and integration with other organizational tools to enhance functionality and scalability.

In conclusion, the project demonstrates a practical and effective solution for improving knowledge management within an organization. It highlights the importance of structured systems in enhancing productivity and ensuring long-term accessibility of information.

9.1 Future Enhancements

While the current system successfully meets the basic requirements of an internal knowledge-sharing platform, there are several areas where further improvements can be made to enhance functionality, performance, and user experience.

One of the key enhancements could be the implementation of an **advanced search mechanism**. This would allow users to find relevant questions more efficiently using filters, keywords, and ranking algorithms based on relevance.

Another important improvement is the addition of a **recommendation system**. Based on user activity and interests, the platform can suggest relevant questions or topics, making the system more intelligent and user-friendly.

The system can also be enhanced by introducing a **notification feature**, where users receive updates when their questions are answered or when there is activity on posts they are following. This would improve user engagement and interaction.

From a scalability perspective, the platform can be deployed on **cloud infrastructure**, enabling it to handle a larger number of users and data more efficiently. This would make the system suitable for enterprise-level usage.

Security can be further strengthened by implementing advanced features such as:

- Role-based access control
- Data encryption
- Secure authentication mechanisms

Finally, future versions of the system could include **analytics and reporting features**, allowing administrators to track user activity, identify popular topics, and gain insights into knowledge usage patterns.

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APPENDIX

i. System Specifications

Hardware Configuration (Development & Testing)

- Laptop/Desktop: Intel Core i5 or equivalent
- RAM: 8 GB
- Operating System: Windows 10/11 or Linux
- Storage: Minimum 256 GB

Software Environment

- Frontend: HTML, CSS, JavaScript
- Backend: Node.js, Express.js
- Database: MySQL
- Development Tool: Visual Studio Code
- Browser: Google Chrome

This setup was used to develop and test the Internal Q&A Knowledge Sharing Platform in a local environment.

ii. Project Implementation Details

- **Project Title:** Internal Q&A Knowledge Sharing Platform
- **Internship Organization:** Bharat Electronics Limited (BEL), PDIC (Product Development and Innovation Center)
- **Internship ID:** PT-717
- **Development Approach:** Full-stack web application
- **Execution Environment:** Localhost server using Node.js

iv. Dataset and Data Handling

Since the project is application-based, no external dataset was used.

Internal Data Used

- User data (ID, username, credentials)

- Questions (title, description, tags, timestamps)
- Answers (linked to questions)
- Voting data

Storage Format

- Structured relational tables in MySQL database

Sample Data

- Dummy/test data created to simulate real user interactions
- Multiple scenarios tested including:
    - Multiple users
    - Multiple questions and answers
    - Voting and answer selection

v. Project Repository / Implementation Files

- **GitHub Repository:** https://github.com/Likith1804/Embedded-Systems

- **Execution Platform:** Visual Studio Code Terminal / Localhost

- **Database Access:** MySQL Workbench / CLI

vi. Sample Screens / Outputs

- **Fig. A.1:** Log-in Page
- **Fig. A.2** Ask New Question Interface
- **Fig. A.3:** Question Posting Interface
- **Fig. A.4:** Home/Dashboard with voting Page
- **Fig. A.5:** Display Question Interface

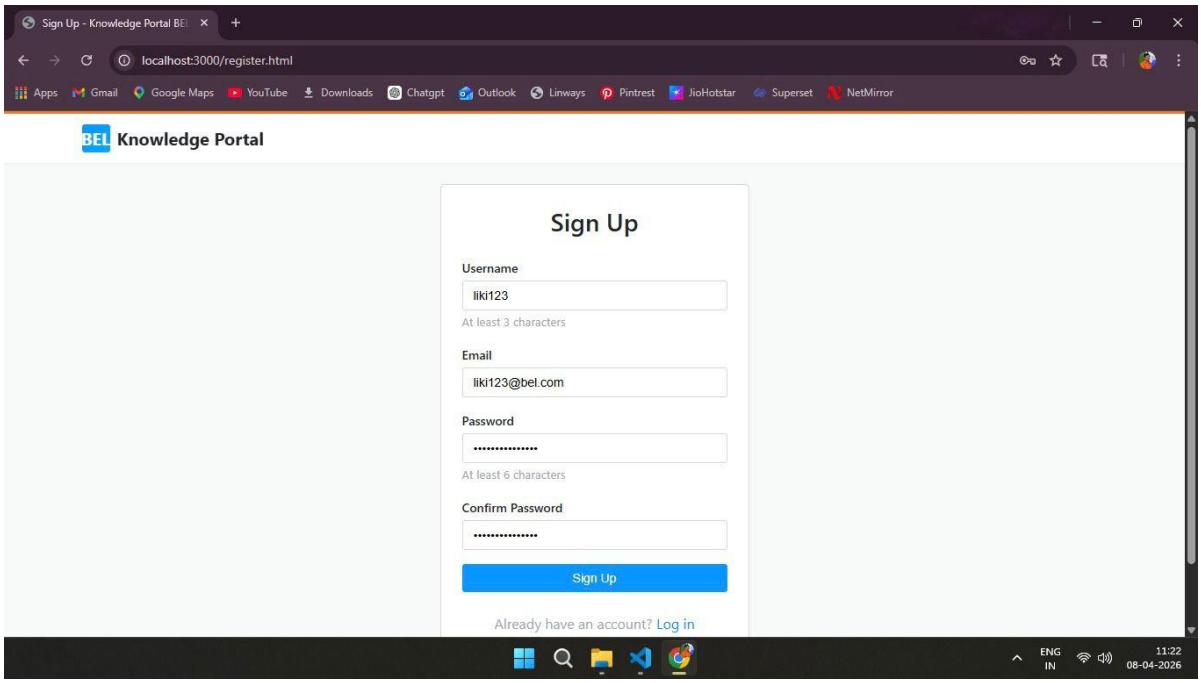

Figure A.1 Log-in Page

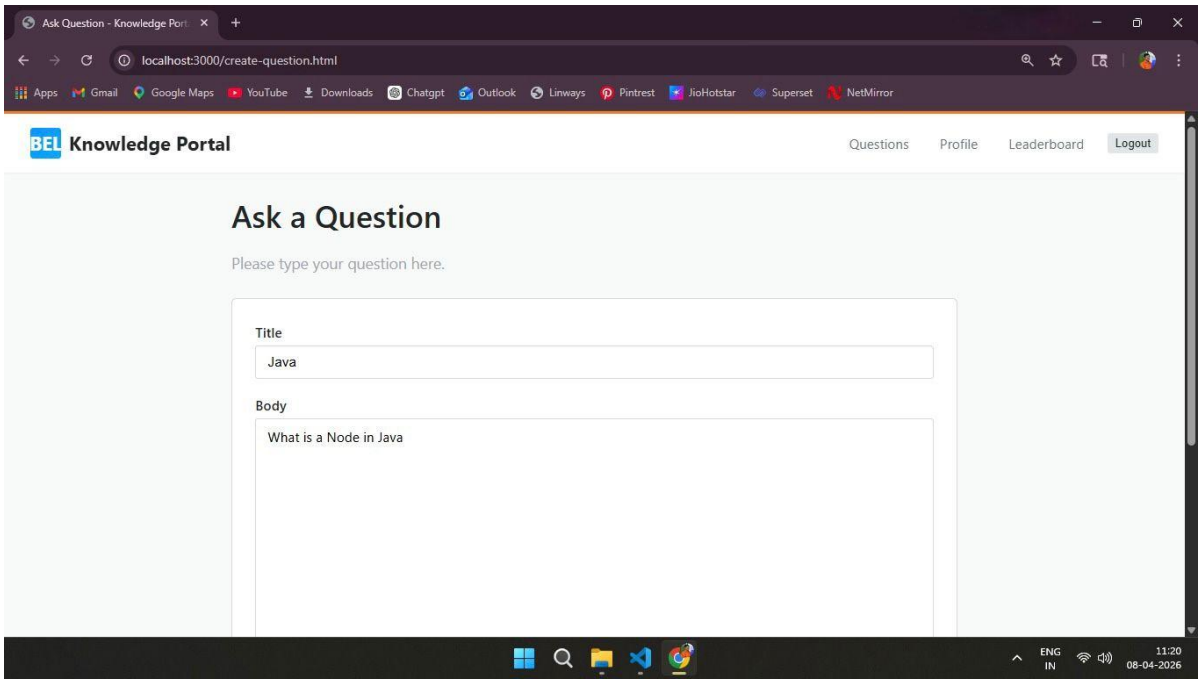

Figure A.2 Ask New Question Interface

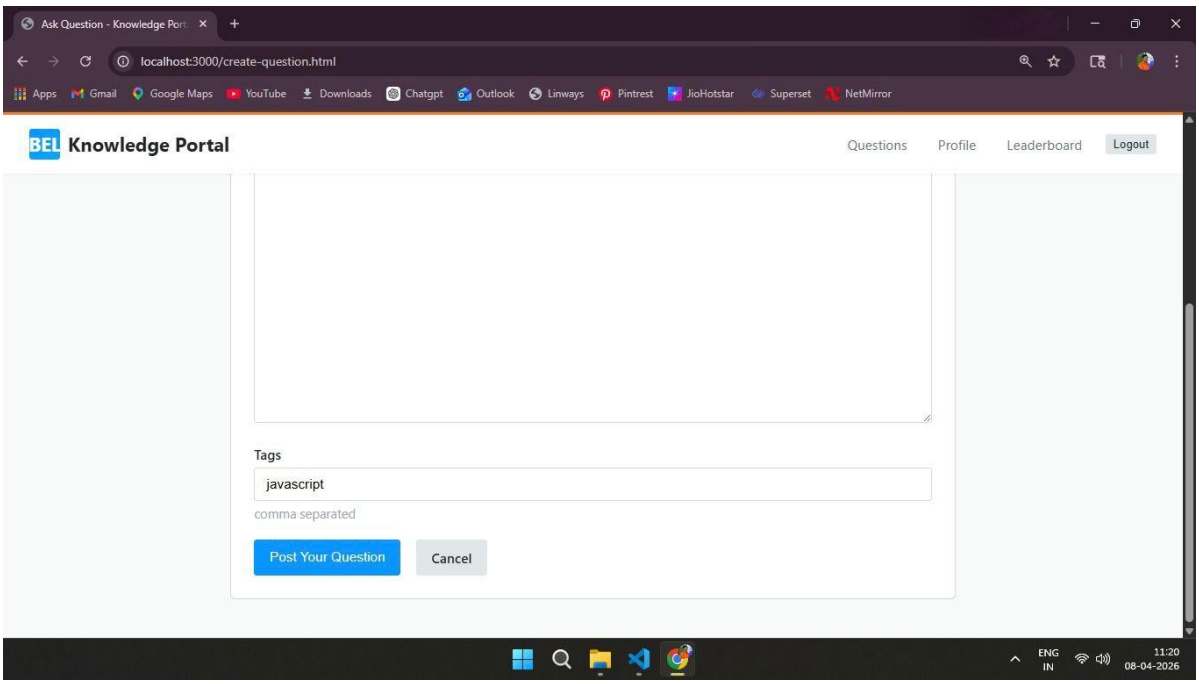

Figure A.3 Question Posting Interface

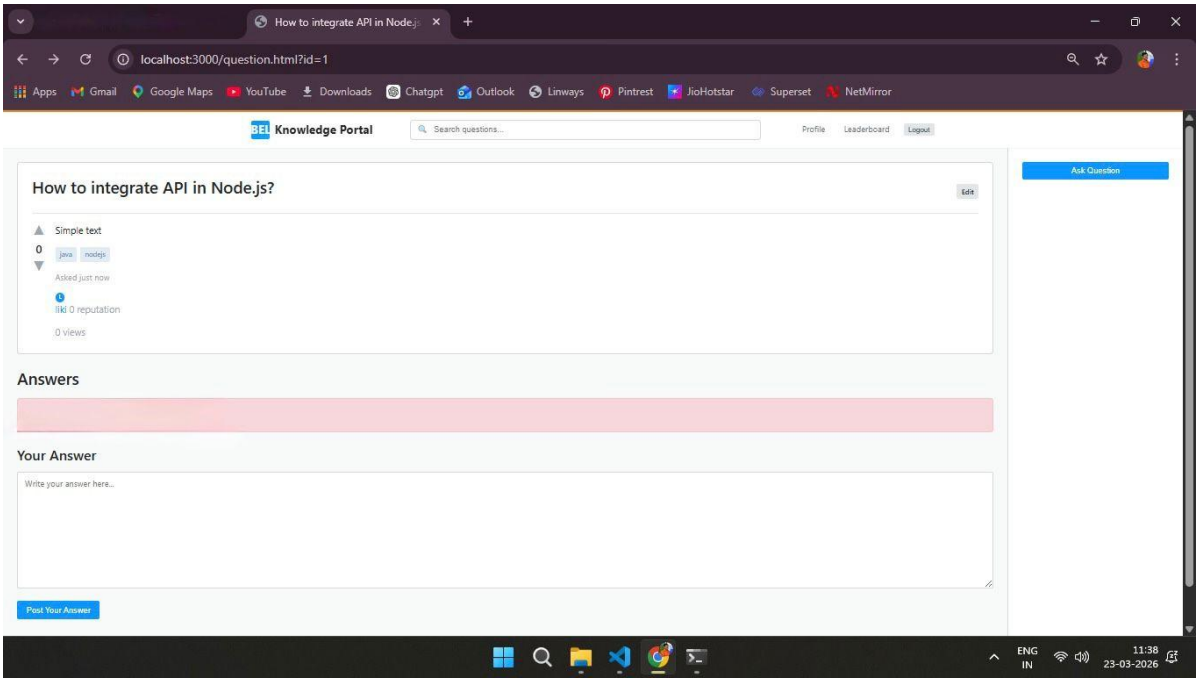

Figure A .4 Home/Dashboard with voting Page

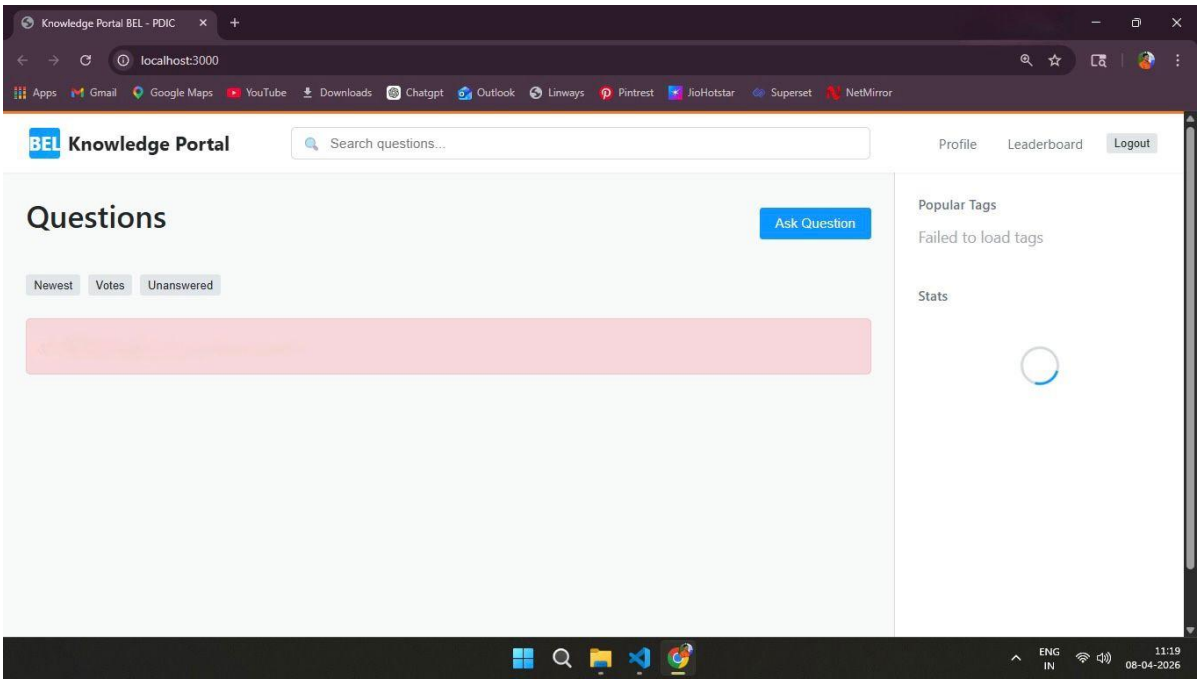

Figure A.5 Display Question Interface

viii. Additional Notes

- The system was developed and tested in a controlled local environment
- No external APIs were required for core functionality
- The platform is scalable and can be extended for enterprise-level deployment